

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)	Cons of energy correctly applied (or implied) i.e. $0.20 \text{ (ecf)} = E_k + E_p$ (at top of loop) (1) Substitution e.g. $0.20 = \frac{1}{2} \times (30 \times 10^{-3}) \times v^2 + (30 \times 10^{-3}) \times 9.81 \times 0.50$ (1) Award two marks if this seen and first mark not already awarded. $v = 1.8[8 \text{ m s}^{-1}]$ accept 1.9 m s^{-1} (1) Award 1 mark for 3.7 m s^{-1} (omitting GPE) Award 3 marks for the following: $\frac{1}{2}mv^2 = 0.2$ leading to $3.65 \text{ [m s}^{-1}]$ (1) can be implied by correct answer Correct application of $v^2 = u^2 + 2as$ i.e. $v^2 = 3.65^2 - 2 \times 9.81 \times 0.5$ (1) Correct answer = $1.87 \text{ [m s}^{-1}]$ (1)		3		3	3	
		(ii)	$5\% \times 0.2 = 0.01 \text{ ecf}$ (1) d correct method: $(\pi \times 0.25) + 0.5 = 1.28[5 \text{ m}]$ (1) $F = \frac{0.01}{1.285} = 0.0078 \text{ [N]}$ ecf on calculated value of d and calculated value of E (1) Accept answers based on 0.9 m or 0.5 m i.e. $d = 1.68[5 \text{ m}]$ so $F = 0.006 \text{ [N]}$ so all 3 marks could be awarded	1	1 1		3	3	
			Question 2 total	3	7	0	10	9	0